ICT FLUX (ict-trader)

PRODUCTION SYSTEM

A deterministic, rules-first algorithmic trading engine and web trading desk that executes institutional ICT (Inner Circle Trader) time-and-price market structure models across forex, commodities, and synthetic indices with strict mathematical risk governance and multi-platform execution.

`Python 3.11+ Core` `MetaTrader 5 & MT4 Bridge` `Deriv WebSockets (~60 Markets)` `Deterministic State Machine (0 ML)` `SQLite Tamper-Proof Audit Journal`

THE PROBLEM IT SOLVES — Eliminating emotional bias and execution failure

⚠️ Emotional & Discretionary Execution

Retail financial traders routinely incur catastrophic drawdowns due to fear, greed, revenge trading, and hesitation. Without algorithmic enforcement, human discipline breaks down during high-volatility sessions.

📐 Subjective Teaching vs. Verifiable Rules

Institutional "smart money" and ICT concepts are widely taught with hindsight cherry-picking. This project translates ambiguous concepts into rigorous, deterministic state machines with verifiable rule IDs.

🔴 Inadequate Risk & Capital Protection

Traders frequently blow accounts by over-leveraging and holding through high-impact news. The engine enforces hard equity loss floors, consecutive-loss cool-offs, and dynamic risk reduction schedules that no logic can override.

🔄 Fragmented Analysis & Broker Disconnect

Screening 60+ markets, calculating session killzones, monitoring higher-timeframe draw, and executing orders across MT4, MT5, or Deriv is manually prohibitive. The system unifies this into a single continuous pipeline.

KEY FEATURES — Verifiable implementations from the codebase

Deterministic Strategy Engines

`ict2022.py · strategy.py`

Step-by-step state machines enforcing ICT rules: Killzones (NY AM/PM, London) → Liquidity Sweep (PDH/PDL, Asia Range) → Market Structure Shift (displacement body multiplier) → Fair Value Gap (FVG) limit entry → Hard Stop beyond sweep extreme → Target at opposing liquidity pool (minimum 1.5R–2R).

Institutional Risk Governance

`risk.py`

Mathematical equity guardrails: Daily and weekly maximum loss floors, automated consecutive-loss circuit breakers (halting trades until session rollover), dynamic risk reduction schedule (halving risk following losing streaks), and small/cent-account compounding sizing.

100-Point ICT Judgement Matrix

`advisor.py · aplus.py`

Scores trade quality out of 100 based on higher-timeframe draw agreement, premium/discount dealing range location, displacement vigor, and path obstacles. Applies mandatory 3-gate A+ checklist before issuing unambiguous calls: *PULL THE TRIGGER*, *SIZE DOWN*, or *STAND DOWN*.

Real-Time 60-Market Scanner

`scanner.py · api/scan.py`

Concurrent scanner streaming 5-minute candles via Deriv public WebSockets and Yahoo Finance across forex majors, metals, indices, crypto, and 24/7 synthetics (Volatility/Crash/Boom). Automatically scales threshold parameters to each market's recent ATR range.

Macroeconomic News Blackout Filter

`news.py`

Ingests economic calendar data and enforces automatic trading halts 30 minutes before and after high-impact currency news events (NF-01). Automatically cancels resting limits and flattens open exposures before volatile releases.

Multi-Platform Execution Bridges

`broker/mt5.py · mql4/ICTBridge.mq4`

Native Python execution for MetaTrader 5 terminals, file-based IPC bridge with custom MQL4 Expert Advisor for MetaTrader 4 (featuring instant stop-loss attachment and auto-close protection for naked positions), and direct Deriv WebSocket auto-trading.

TECH STACK & INFRASTRUCTURE — Standardized, robust, zero-bloat engineering

CORE & MODELING
- **Python 3.11+** Stdlib
- `dataclasses` state machines
- `sqlite3` audit ledger
- `math` & `statistics`
- Zero opaque ML black-boxes

FRONTEND & DESK
- **Vanilla JS & HTML5**
- Lightweight Charts (TradingView)
- **Three.js** 3D market visualization
- Service Worker PWA (`sw.js`)
- Dual Dark/Light responsive UI

EXECUTION BRIDGES
- **MetaTrader 5** Python API
- **MetaTrader 4** MQL4 Bridge EA
- **Deriv WebSocket API**
- Heartbeat health monitor
- Emergency STOP file trigger

CLOUD & TELEMETRY
- **Vercel Serverless** Python
- **Telegram Bot API** alerts
- `ntfy.sh` push dispatch
- GitHub Actions cron runners
- `journal.sqlite` persistence

HOW IT WORKS — Plain-English architectural flow

STEP 01
Ingestion & Bias
Market data flows in per candle close. `narrative.py` evaluates higher-timeframe weekly/daily bias and tracks unpurged liquidity pools (prior day/week highs & lows).

STEP 02
Tactical Triggers
Inside active NY killzones, `ict2022.py` detects reference level sweeps. A displacement candle shifting market structure arms a resting limit in the Fair Value Gap.

STEP 03
Scoring & Risk Gate
The 100-pt matrix assesses confluence. `risk.py` validates daily loss floors, news blackout status, and sizes the order so dollar risk exactly matches the loss budget.

STEP 04
Execution & Audit
Order is dispatched to MT5, MT4 EA, or Deriv with mandatory stop loss. Every trigger, rule evaluation, and fill is committed to SQLite and broadcast via push alerts.

CURRENT STATUS & NEXT STEPS — Verifiable progress and honest engineering roadmap

✅ Built & Verified in Codebase
- • **Complete deterministic core:** ICT 2022 Mentorship, Turtle Soup sweep-and-reverse, and Silver Bullet state machines.
- • **Full risk engine:** Equity floors, trade caps, news blackouts, and dynamic position sizing verified across 50+ unit tests.
- • **Multi-platform execution:** Working MT5 Python connector, MT4 MQL4 bridge EA, and Deriv WebSocket auto-trader.
- • **Live web desk & scanner:** PWA trading desk with session clocks, interactive charts, 60-pair scanner, and Telegram alerts.
- • **Walk-forward validation:** Realistic backtester with commissions, spread, stop-first execution, and random-walk noise canary.

⏳ In Progress & Planned Roadmap
- • **Automated rule extraction:** Parsing mentor video transcripts to replace baseline ICT defaults with client-specific nuances.
- • **Journal reconciliation:** Automated ingestion and cross-comparison against the trader's personal historical trade journals.
- • **Live telemetry dashboard:** Centralized real-time web monitoring interface querying multi-terminal SQLite databases.
- • **Multi-tenant user billing:** Production auth and payment integration (currently running as a full client-side Pro preview).

RECOMMENDED DEMO SHOWCASE — Open this file during your client video demo

PY

Primary Logic File to Feature: `ict_trader/ict2022.py` **(1,137 lines)**

Why it best demonstrates your work: This file contains the complete ICT 2022 Mentorship state machine. It proves how complex market dynamics are reduced to pure deterministic code: session timing, liquidity sweeps, displacement candle body calculations, fair value gap entries, and rule-tagged decision logging (KZ, SW, MS, FV, SL, TP).